

Tirumala Venkata Durga Prasad Pitani

AI Engineer — Agentic AI — Generative AI — Enterprise Automation

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Professional Summary

AI Engineer with 2+ years of experience specializing in Generative AI, agentic workflows, and enterprise automation. Proven track record of designing and deploying production-ready LLM applications, RAG systems, and intelligent agents on Microsoft Azure. Adept at transforming complex business requirements into scalable, secure AI solutions that drive measurable impact, such as reducing manual processing time by 50% and serving thousands of users with high-accuracy NL-to-SQL capabilities. Recognized innovator with top finishes in multiple enterprise AI hackathons.

Technical Skills

- **Generative AI & Agents:** Azure OpenAI, LangChain, LangGraph, LlamaIndex, ReAct Agents, Tool-Use Agents, Multi-Agent Orchestration, Prompt Engineering, Structured Outputs
- **RAG & Search:** FAISS, ChromaDB, Pinecone, Embeddings, Vector Search, Semantic Chunking, Hybrid Retrieval, Grounding, NL-to-SQL
- **Programming & APIs:** Python (Async), SQL, Pandas, NumPy, FastAPI, Django REST Framework, JWT, RBAC
- **Cloud & Data Platforms:** Microsoft Azure, Azure Fabric, Azure Data Factory, Azure ML Studio, Databricks, PySpark
- **MLOps & DevOps:** MLflow, Scikit-Learn, XGBoost, Docker, CI/CD (Azure DevOps), Automated Testing

Professional Experience

Data Engineer / AI Engineer

NTT DATA Business Solutions

October 2024 – Present

Bengaluru, India

- Designed and deployed an **LLM-powered enterprise analytics copilot** using LangChain and FAISS, enabling natural-language queries against governed lakehouse data with **85% NL-to-SQL accuracy**, reducing manual analyst effort by **50%**.
- Developed reusable AI components (prompt construction, retrieval, SQL generation, validation) to accelerate enterprise AI adoption, supporting maintainable and scalable multi-agent workflows.
- Built AI-ready data infrastructure on **Azure Fabric** using PySpark, processing 50+ GB of daily supply-chain data through Medallion Architecture; optimized Delta Lake workloads, improving query performance by **45%**.
- Engineered 15+ automated ETL/ELT pipelines in **Azure Data Factory** with dynamic parameterization and data-quality validation, achieving **99.8% on-time delivery** for critical business reporting.

Data Science Intern

Viai Solutions Pvt Ltd

July 2024 – August 2024

Hyderabad, India

- Developed automated Python data-ingestion workflows (Selenium, BeautifulSoup) to extract and transform unstructured web data into Parquet datasets, reducing manual collection effort by **40%**.
- Processed and validated 100K+ student records using PySpark, improving data quality from **72% to 96%**.

Python Developer

Dhanush Infotech Pvt Ltd

January 2024 – June 2024

Hyderabad, India

- Developed scalable Django REST Framework microservices with JWT authentication and role-based access control, supporting 10K+ daily API requests with **99.9% uptime**.
- Optimized SQL schemas and query execution plans, improving API response times by **20%** and reducing database CPU utilization by **30%**.

Key Projects

Enterprise AI Analytics Copilot

LangChain, FAISS, Azure Fabric, PySpark, SQL

- Built a GenAI copilot combining RAG, vector retrieval, and context grounding to enable secure, self-service natural language querying of curated Gold-layer enterprise datasets.
- Implemented validation-oriented SQL generation workflows to ensure response accuracy and reliability.

Cross-Sell Prediction System

Python, Scikit-Learn, Pandas, MLflow

- Engineered a machine learning pipeline to predict customer cross-sell opportunities, utilizing Logistic Regression and Support Vector Machines to analyze customer behavior patterns.
- Implemented comprehensive data preprocessing, feature engineering, and model evaluation metrics (ROC-AUC), deploying the solution to identify high-probability conversion targets and optimize marketing workflows.

Achievements & Certifications

- **1st Place** – Business AI Challenge 2026 (Cohort 1); **3rd Position** – JouleXet 2025 AI Hackathon
- **Certifications:** Microsoft Certified: Fabric Data Engineer Associate (DP-700), Azure AI Fundamentals (AI-900)
- **Publication:** "CNN-based Industrial Defect Detection System" (94% detection accuracy)

Education

B.Tech (Computer Science and Engineering)

IIT Manipur

2020–2024